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A conceptual model of preferential flow systems in forested hillslopes: evidence of self-organization

Roy C. Sidle,[1]* Shoji Noguchi,[2] Yoshio Tsuboyama[2] and Karin Laursen[3]

[1] *Department of Geography, National University of Singapore, I Arts Link, Singapore 117570*
[2] *Forestry and Forest Products Research Institute, PO Box 16, Tsukuba Norin Kenkyu, Ibaraki 305-8687, Japan*
[3] *Environmental Technology Institute, 18 Nanyang Dr., Singapore 637723*

Abstract:

Preferential flow paths are known to be important conduits of subsurface stormflow in forest hillslopes. Earlier research on preferential flow paths focused on vertical transport; however, lateral transport is also evident in steep forested slopes underlain by bedrock or till. Macropores consisting of decayed and live roots, subsurface erosion, surface bedrock fractures, and animal burrows form the basis of a 'backbone' for lateral preferential flow in such sites. Evidence from field studies in Japan indicates that although individual macropore segments are generally $<0{\cdot}5$ m in length, they have a tendency to self-organize into larger preferential flow systems as sites become wetter. Staining tests show clear evidence of interconnected macropore flow segments, including: flow within decayed root channels and subsurface erosion cavities; flow in small depressions of the bedrock substrate; fracture flow in weathered bedrock; exchange between macropores and mesopores; and flow at the organic horizon–mineral soil interface and in buried pockets of organic material and loose soil. Here we develop a three-dimensional model for preferential flow systems based on distributed attributes of macropores and potential connecting nodes (e.g. zones of loose soil and buried organic matter). We postulate that the spatially variable and non-linear preferential flow response observed at our Japan field site, as well as at other sites, is attributed to discrete segments of macropores connecting at various nodes within the regolith. Each node is activated by local soil water conditions and is influenced strongly by soil depth, permeability, pore size, organic matter distribution, surface and substrate topography, and possibly momentum dissipation. This study represents the first attempt to characterize the spatially distributed nature of preferential flow paths at the hillslope scale and presents strong evidence that these networks exhibit complex system behaviour. Copyright © 2001 John Wiley & Sons, Ltd.

KEY WORDS macropores; preferential flow; subsurface flow; forest hydrology; hillslope hydrology; complex systems; root channels; conceptual models; fracture flow

INTRODUCTION

The contribution of preferential flow paths in soils and regoliths to the rapid transport of water and solutes has been recognized for some time. Though much focus has been placed on vertical transport of pollutants in agricultural and urban sites (e.g. Thomas and Phillips, 1979; Edwards *et al*., 1988; Rowe and Booker, 1990; McKay *et al*., 1993; Elliot *et al*., 1998; McMurry *et al*., 1998; Broholm *et al*., 2000), a few earlier studies alluded to the complex macropore networks inherent in forest soils (Whipkey, 1969; Aubertin, 1971; Sidle *et al*., 1977; Sidle and Kardos, 1979). Later research in steep forested watersheds identified lateral subsurface flow through macropore systems as a potential direct link to rapid runoff response in forest streams (Mosley, 1979, 1982; Tsukamoto and Ohta, 1988; Kitahara *et al*., 1988, 1994). Results from some of the earlier small-scale studies were challenged by a group of New Zealand researchers who observed isotopically 'old' water dominating stormflow runoff (Pearce *et al*., 1986; Sklash *et al*., 1986). These studies incorrectly associated

* Correspondence to: R. C. Sidle, Department of Geography, National University of Singapore, I Arts Link, Singapore 117570.
E-mail: geosrc@nus.edu.sg

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low levels of isotopically ‘new’ water (or rainwater) in storm runoff with small preferential flow contributions to storm runoff. Later studies at the same site refined the explanation (McDonnell, 1990), but tangible evidence related to specific flow paths remains elusive.

More recent research at Hitachi Ohta Experimental Watershed in Japan suggests that preferential flow systems are dynamic and that the extent of their hydrologic activity is influenced strongly by antecedent moisture (Tsuboyama *et al*., 1994; Sidle *et al*., 1995, 2000). Related studies indicate individual macropores that make up preferential flow systems are typically very short ($<0{\cdot}5$ m in length); however, they are able to ‘connect’ over relatively long slope distances due to various mechanisms (Noguchi *et al*., 1999). Sidle *et al*. (2000) suggest such linkages might result from complex system behaviour where nodes (conditioned by antecedent moisture) form the critical system links. It is important to contrast the type of preferential flow network at Hitachi Ohta with sites where large and continuous subsurface soil pipes develop. Such large-scale interconnected pipe systems have only been documented in a few instances, largely in the UK (e.g. Jones, 1971; Smart and Wilson, 1984; Chappell *et al*., 1990) and in certain dry regions (e.g. Fletcher *et al*., 1954). Though several studies in forested sites have reported flow from large pipes (e.g. Roberge and Plamondon, 1987; Kitahara *et al*., 1988, 1994; Tsukamoto and Ohta, 1988; Uchida *et al*., 1999), it is not clear whether these pipes are continuous over long slope distances.

Understanding the spatial and temporal dimensions of preferential flow paths has strong implications for hydrologic modelling (Anderson and Rogers, 1987; Chappell and Ternan, 1992) and other practical applications. It is important to know when and where, within a catchment, soils are hydrologically active. Such knowledge is needed to schedule logging, grazing rotations, application of pesticides and chemicals, and road construction. Additionally, this knowledge can help determine where within the catchment such activities can be conducted safely. Cumulative effects of dispersed land management practices on water resources can then be better understood (Megahan *et al*., 1991; Sidle and Hornbeck, 1991; Dixon and Montz, 1995). Indeed, research on subsurface flow pathways was listed as a major priority in the USDA initiative on ‘Water Quality: Cumulative Effects from Forests and Rangelands’ (Sidle *et al*., 1989).

Specification of preferential flow paths in catchments is critical to assessing the assumptions and inferences in many tracer and natural isotope investigations. From the rather simplistic two-component, end member mixing analysis (EMMA), which differentiates ratios of $^{18}O/^{16}O$ in rain water and soil water (Sklash and Farvolden, 1979; Pearce *et al*., 1986; Cey *et al*., 1998), to the more complex multi-component EMMA models that assign chemical signatures to additional system components (DeWalle *et al*., 1988; McDonnell *et al*., 1991; Bazemore *et al*., 1994; Elsenbeer *et al*., 1995), certain inherent assumptions greatly restrict any inferences related to temporal and spatial attributes of preferential flow paths. The most important assumption in EMMA models is that the signature of each end member is conservative (i.e. no mixing among sources or compartments and no changes of signatures within a compartment). Even though most investigators employing EMMA clearly state these assumptions, in many cases they do not recognize their limitations in terms of inferences, i.e. with respect to flow path designation. The recent literature abounds with examples where various types of EMMA applications have inferred flow paths (e.g. Robson and Neal, 1990; Pearce, 1990; McDonnell *et al*., 1991; Bazemore *et al*., 1994; Brown *et al*., 1999).

In this paper, we outline a conceptual approach to capture the complex behaviour of preferential flow systems in a forest hillslope using data from the Hitachi Ohta Experimental Watershed in Japan. Here we assume that slope length greatly exceeds soil depth, infiltration capacities are generally higher than incident rainfall intensities, and the relatively shallow soils are underlain by bedrock or till of lower permeability (but not impermeable). These conditions are common in steep terrain of temperate and tropical forest environments; thus the general approach should apply in many areas. Developing such a model that simulates the spatial and temporal dynamics of preferential flow systems is an important contribution to our understanding of hillslope and catchment hydrology, particularly stormflow generation and transport of solutes to streams (Anderson and Rogers, 1987; Chappell and Ternan, 1992; Sidle *et al*., 2000).

PREFERENTIAL FLOW SYSTEM COMPONENTS

Detailed investigations at a hillslope site in Hitachi Ohta indicate that individual macropores in the soil rarely exceed 0·5 m in length and never exceed 0·62 m (Noguchi *et al*., 1999). Thus, the downslope continuity of subsurface preferential flow paths cannot be attributed to individual macropores. Rather, the continuous preferential flow paths that have been evidenced by both hydrometric measurements (Sidle *et al*., 1995) and tracer tests (Tsuboyama *et al*., 1994) and observed directly from staining tests (Noguchi *et al*., 1999) at the Hitachi Ohta hillslope can only be explained by a complex three-dimensional linkage of macropores in the soil and substrate that facilitate an upslope extension of the preferential flow system. Although other studies in forested sites have inferred the existence of long soil pipes (Roberge and Plamondon, 1987; Kitahara *et al*., 1988, 1994; Uchida *et al*., 1999), excavations at other forested sites revealed that the continuity of individual pipes was at most several metres (Tsukamoto and Ohta, 1988). Sidle *et al*. (2000) propose that such a dynamic preferential flow system is linked by a series of 'nodes' of connectivity that can be conditioned by different levels of antecedent moisture. The nodes may include physical interconnection of short macropore segments, buried pockets of organic matter or loose soil, and direct interaction with a lithic boundary (including fractures).

To quantify the three-dimensional connectivity of preferential flow systems, we utilize detailed data from numerous soil pits within the Hitachi Ohta catchment (Noguchi *et al*., 1997), macropore and preferential flow path staining tests in an excavated slope section (Noguchi *et al*., 1999) and measurements of fractures and cleavage planes in the dominant schist bedrock. Inclusion of these extensive data sets will provide the basis for a realistic description of three-dimensional preferential flow systems. Saturated hydraulic conductivity values based on soil cores collected throughout the Hitachi Ohta catchment ranged from $2{\cdot}94 \times 10^{-5}$ to $7{\cdot}37 \times 10^{-4}$ ms^{-1} (Noguchi *et al*., 1997). Though these data may be useful for comparative purposes, tracer tests (Tsuboyama *et al*., 1994) and subsurface flow investigations during natural storms (Sidle *et al*., 1995, 2000) indicate that such values would underestimate the flux of lateral flow at the hillslope scale. Noguchi *et al*. (1997, 1999) present additional data on soil physical properties at Hitachi Ohta.

Soil macropores

Soil macropores with diameters >2 mm and lengths >2 cm were mapped in nine soil pits at the Hitachi Ohta site (Figure 1). Macropores were largely formed by decayed root channels and subsurface erosion or the interaction of these two features (Noguchi *et al*., 1997). Additionally, one of the soil pits (pit F, see Figure 1) was carefully excavated upslope in 10 cm 'slices' for a total slope distance of 2 m (Noguchi *et al*., 1999). Thus, information on macropore properties was collected for an additional 20 soil pits or slices.

Density. Macropore densities were calculated for both the topsoil (O and A horizons) and subsoil (B and C horizons) using entire soil pits as the sampling unit to ensure a representative cross-sectional area of substrate. The width of the soil pits varied from 1·00 to 1·85 m. All 29 soil pits (or slices of pits) were included to develop a probability density function (PDF) for macropore density at each depth interval (Figure 2). Macropore density in the topsoil ranged from 0 to 168·1 macropores m^{-2}, with an average value of 74·9 macropores m^{-2}. The density in the subsoil ranged from 4·9 to 84 macropores m^{-2}, with an average of 30·6 macropores m^{-2}. Macropore density in the subsoil was $\leq$50 macropores m^{-2} in more than 86% of the cases (Figure 2b). The density in topsoil layers was more evenly distributed throughout the full range from 0 to 180 macropores m^{-2} (Figure 2a). For forest soils with significant organic-rich horizons, it appears useful to separate the densities and other characteristics of macropores between the topsoil and subsoil layers. No attempt was made to account for macropores that began in the topsoil and entered into the subsoil (or *vice versa*); they were simply tabulated according to their location in a particular two-dimensional pit face. However, macropore orientation determines the potential intersection or interaction of these pathways between the topsoil and subsoil.

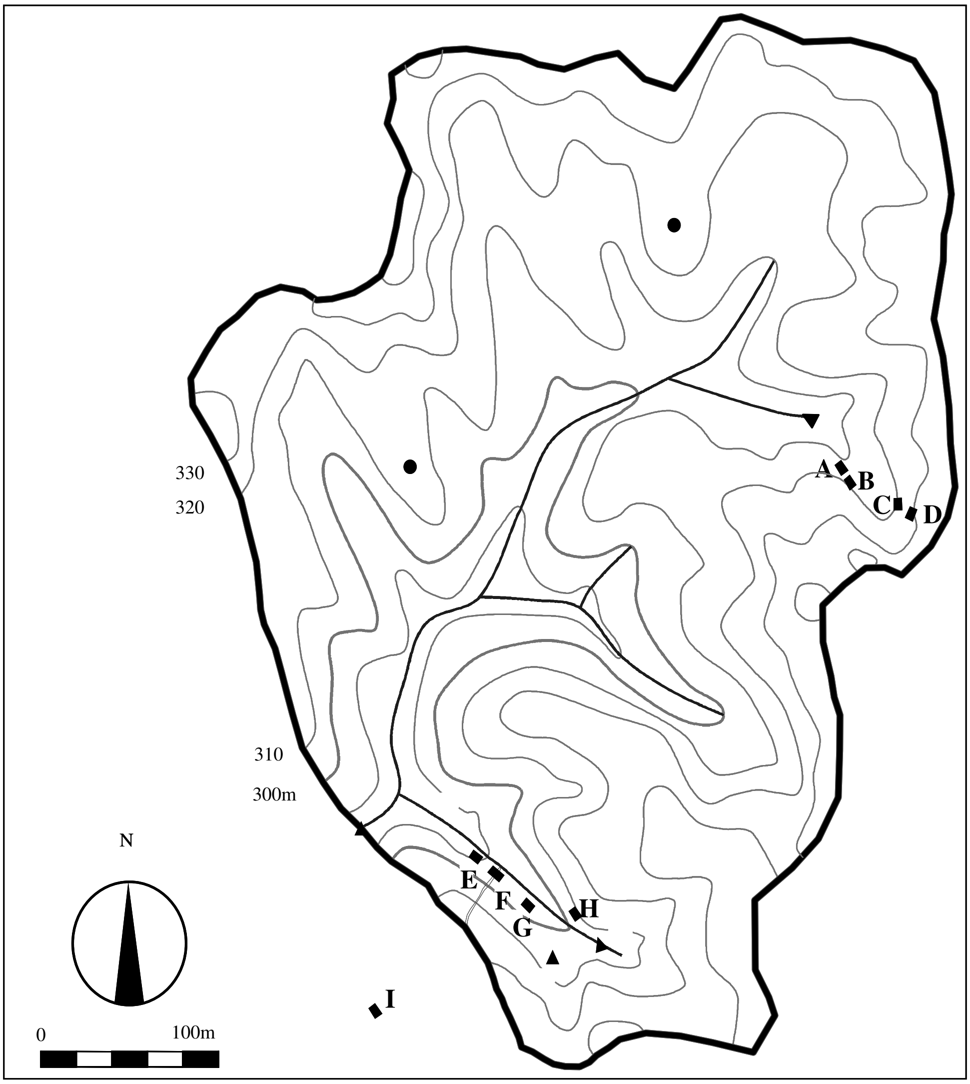

Figure 1. Map showing the locations of soil pits within the vicinity of the Hitachi Ohta Watershed. Pit F had 20 sequential slices excavated at 8 cm intervals. Climate stations • and gauging weir locations ▲ are shown

Distribution. The spatial distribution of macropores at individual soil pits can be described by Morisita's (1959) I_δ index

$$I_\delta = q\frac{\sum_{i=1}^{q} n_i(n_i - 1)}{N(N-1)} \tag{1}$$

where n_i is the number of macropores in each section ($i = 1, 2, \ldots, q$), q is the number of sections of a designated grid size in each pit face, and N is the total number of macropores in all grid sections

$$N = \sum_{i=1}^{q} n_i \tag{2}$$

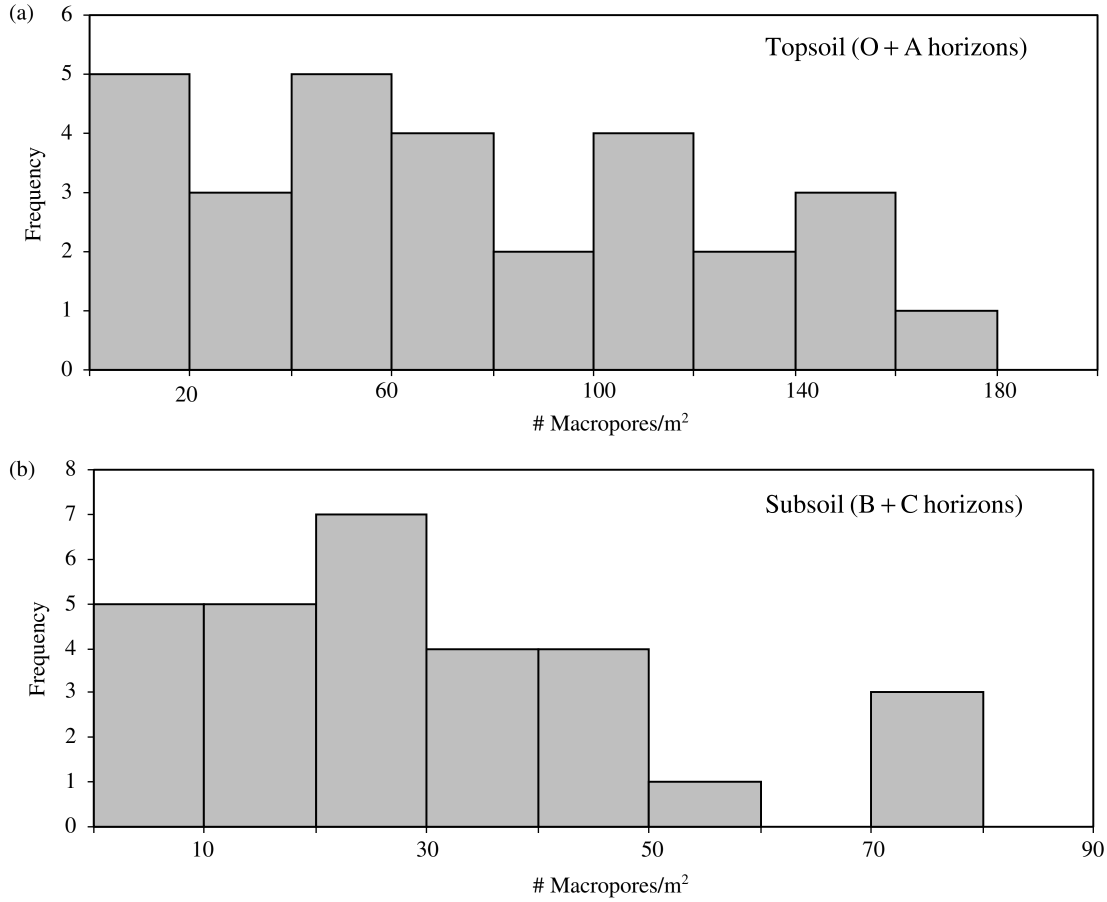

Figure 2. PDFs for macropore density (macropores m^{-2}) in the topsoil (a) and subsoil (b) of 29 soil pits or slices in the Hitachi Ohta Experimental Watershed

Practically, the smallest grid size selected (in our example at Hitachi Ohta: 10 cm $\times$ 10 cm) is constrained by the density of macropores, i.e. most grids should contain one or more macropores. The largest grid selected (in our case 80 cm $\times$ 80 cm) is primarily constrained by the soil depth, i.e. the grid size should be no greater than the soil depth or pit width. Thus, the range of grid sizes selected for any particular site would depend on macropore density, soil depth, and pit width. Morisita (1959) developed characteristic relationships between I_δ and grid size that are useful for determining the evenness of the spatial distribution of individuals in a population. Experimental data can be compared qualitatively against these theoretical relationships to ascertain which distribution (random, uniform, aggregated with small clumps, aggregated with large clumps) best fits the data. Noguchi *et al*. (1999) showed these characteristic relationships between I_δ and grid size and found that six soil slices in Hitachi Ohta hillslope had sufficient macropores (minimum of 30) to be analysed by this statistical methodology. These macropores were aggregated in large clumps and the intra-clump distribution was rather uniform according to Morisita's (1959) relationships. Alternatively, Kitahara *et al*. (1988) observed that larger soil pipes in gently sloping forest soils in Hokkaido, Japan, were aggregated in small clumps. If enough macropores exist, it is best to characterize their distributions in both the topsoil and subsoil. However, the depth constraints in the topsoil would typically limit such an approach.

Length. Individual macropore length at Hitachi Ohta could only be calculated based on data collected from one pit where the 20 consecutive slices were excavated. The total length for individual macropores was estimated as the sum of the straight-line segments through each soil slice or portion of each slice as determined by spraying coloured powdered chalk into macropore cavities at each excavated pit face (Noguchi *et al.*, 1999). At the hillslope site the soil macropores ranged in length from 2·0 to 61·8 cm, although about 70% of all macropores were discontinuous after one slice ($\sim$8 cm horizontal distance). The PDF for macropore length indicates a skewed distribution, with many macropores in the 2–10 cm length range and very few macropores $>$25 cm (Figure 3). These data emphasize that individual long macropores do not contribute significantly to extended slope drainage.

Tortuosity. The tortuosity of individual macropores can affect the routing of preferential flow as well as the probability for macropores to interact with other preferential flow paths. Tortuosity is defined here as the total macropore length divided by the distance between the endpoints of the macropore. The tortuosity of individual macropores at the excavated hillslope segment in Hitachi Ohta ranged from 1·00 (straight) to 1·52, with a mean of 1·14 (Noguchi *et al.*, 1999). Tortuosity was weakly correlated with macropore length: longer macropores had higher but more variable tortuosity values (Noguchi *et al.*, 1999) (Figure 4). For the purposes of this hillslope site, tortuosity is thus evaluated in three categories of macropore length (Figure 4). For each category, the tortuosity is randomly sampled (to the nearest 0·05) from the mean $\pm$ one standard deviation (SD). For the macropores with lengths of 16·5 to 19·5 cm, the mean was 1·04 and the SD was 0·035. Because tortuosity was calculated only for macropores that penetrated at least two 10 cm slices in the field, these values should be applied cautiously to shorter macropores. Macropores $>$19·5 and $<$40 cm had the following tortuosities: mean = 1·14 and SD = 0·101. The longest ($>$40 cm) macropores had the most variable tortuosities: mean = 1·24 and SD = 0·210.

Studies using soft X-ray techniques detected macropore tortuosities in the ranges of 1·2 to 2·0 (Narioka, 1990) and 1·1 to 1·48 (Mori *et al.*, 1999) for forest soils in Japan. The sample size for these investigations was small (84–125 cm^3). In larger samples (2708 cm^3) of forest soils, Kitahara (1989) found that tortuosities for larger soil pipes (measured by plaster casts) fell within a narrow range of 1·0 to 1·1. Thus, the tortuosity values obtained at Hitachi Ohta appear reasonable for forest soils with complex smaller diameter macropore and can be used to estimate this spatial component in a three-dimensional model.

Diameter. Though macropore diameter is undoubtedly important in dictating the potential interaction or intersection of various macropore segments, the influence of diameter on hydrology of smaller macropores may not be as great as other physical properties, since macropores are typically only partially filled with water.

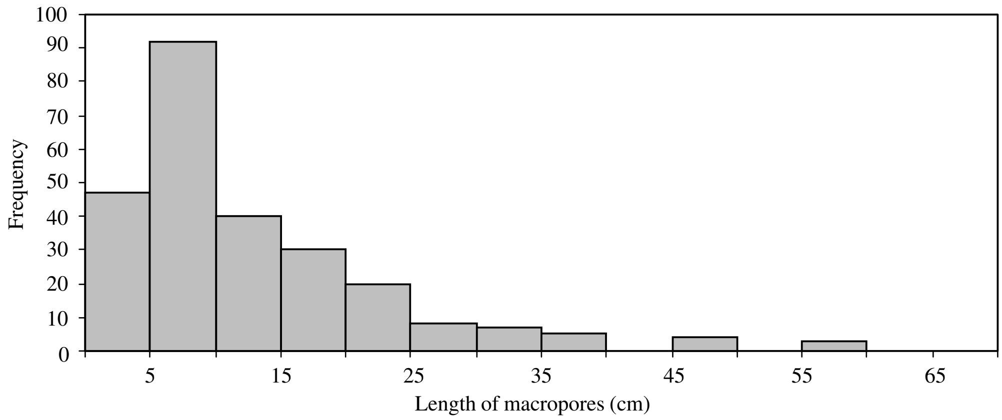

Figure 3. PDF for macropore length derived from 20 consecutive excavated slices in a hillslope at Hitachi Ohta (after Noguchi *et al.* (1999). Reproduced by permission of SSAJ)

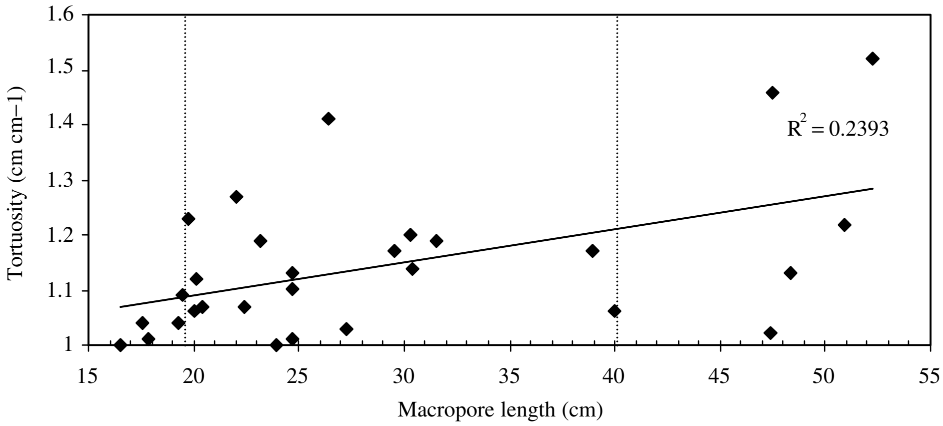

Figure 4. Relationship between macropore tortuosity and macropore length derived from 20 consecutive excavated slices in a hillslope at Hitachi Ohta (after Noguchi *et al*. (1999). Reproduced by permission of SSAJ). Three macropore length categories used for tortuosity simulations are shown by the dashed lines

The diameters of macropores vary in space and time and reflect the interactions among forming processes (e.g. decaying roots, desiccation), expansion (e.g. subsurface hydraulic erosion), maintenance (e.g. excavation by burrowing fauna), and disappearance (e.g. collapse) in macropore systems. Even individual macropores may have varying diameters over relatively short lengths. In the case of individual macropore segments, it is the smallest diameter or bottleneck that regulates maximum flow. Because of these dynamic influences and the high coefficient of variability for macropore diameter (64%) at the Hitachi Ohta site, we evaluated diameter data from all 29 soil pits or slices within the catchment. Mean macropore diameter was higher in the topsoil (19·8 mm) than in the subsoil (13·5 mm). Macropore numbers in the topsoil exhibited a Gaussian distribution across the diameter range of 2–40 mm (Figure 5a). However, for the subsoil, almost half of the macropores were lumped in the 2–10 mm diameter classes (Figure 5b) (Noguchi *et al*., 1997).

Orientation. The orientation of macropores in steep hillslopes may be strongly related to the formative elements, i.e. orientation of root systems and subsurface flow vectors (Noguchi *et al*., 1999). To orient the soil macropores in three dimensions, both vertical gradient (analogous to slope gradient) and planar orientation must be specified. The gradient is zero for macropores that are perpendicular to the pit face (i.e. zero slope); angles in the downward direction are positive, and those in the upward direction are negative (Noguchi *et al*., 1997). Planar orientation is referenced to the pit face, with zero set for the direction of the face and angles in clockwise and counterclockwise directions assigned positive and negative values respectively (Noguchi *et al*., 1997). Because the initial specified gradient and planar orientation used in simulations may exert a major influence on overall macropore orientation as well as the potential for macropores to cross over from the topsoil to subsoil (or *vice versa*), we present PDFs for these two parameters that are: (1) representative of the entire catchment (i.e. data from the nine distributed soil pit faces); and (2) representative of the small-scale variability within a hillslope section (i.e. data from the 20 slices at pit F).

Macropore gradients in the topsoil of the nine distributed pits were 90% negative oblique (Figure 6a), whereas the subsoil had a more uniformly balanced distribution of negative and positive gradients (Figure 6c). The PDFs for the small-scale measurements at multiple slices of a single pit indicated a similar trend (Figure 6b and d). The preponderance of negative oblique gradients in the topsoil likely reflects the orientation of surface roots in this layer (Karizumi, 1979). The planar orientation of macropores in the topsoil was rather evenly distributed at both the large and small scales, although few macropores were nearly perpendicular to the pit face (i.e. ±70–90°) (Figure 7a and b). In the subsoil, about 63% of the large-scale macropores and 66% of the small-scale macropores were clustered in the range from ±0–40° (Figure 7c and d).

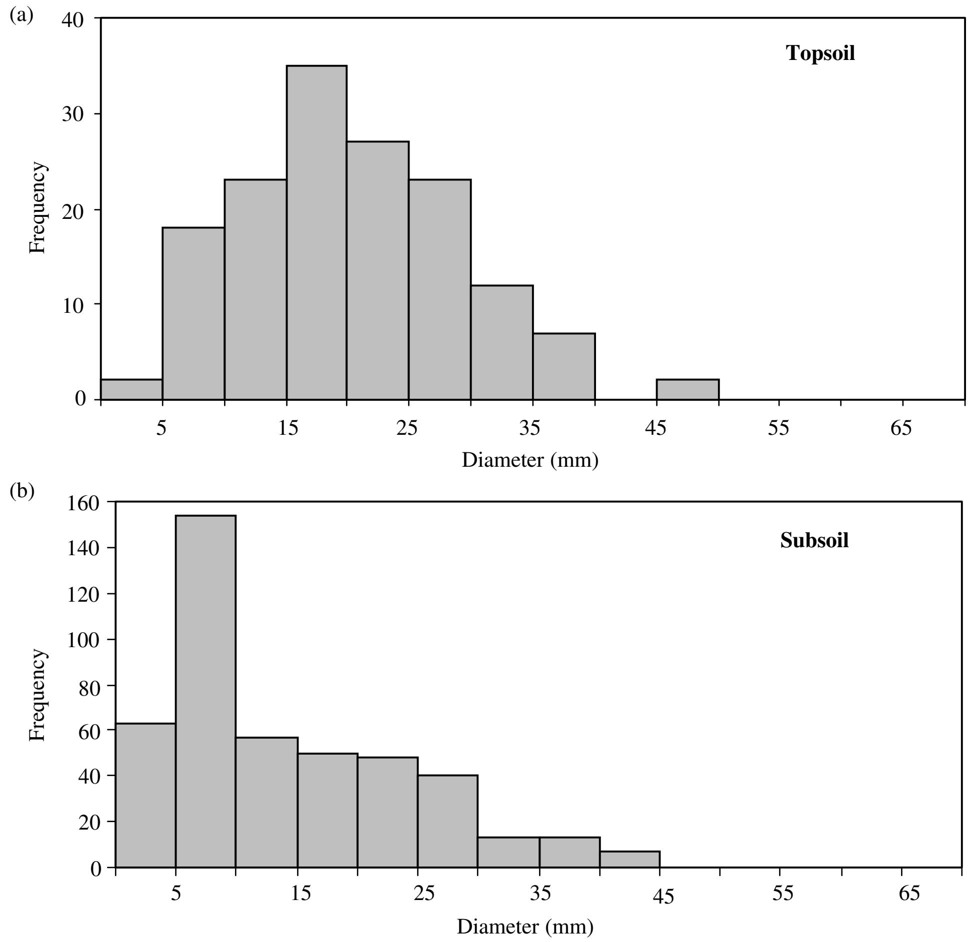

Figure 5. PDFs for macropore diameter in the topsoil (a) and subsoil (b) of 29 soil pits or slices in the Hitachi Ohta Experimental Watershed

Preferential flow paths in bedrock and till

Many forest hillslopes are composed of relatively thin, permeable soil mantles overlying less-permeable bedrock or till. The assumption made for most studies conducted at such sites is that the basal bedrock or till serves as a hydrologic discontinuity and deflects flow parallel to the slope (e.g. Harr, 1977). However, recent field studies in forest soils show that preferential flow exuding from bedrock fractures can exert an important control on subsurface flux (Tsukamoto and Ohta, 1988; Hirose *et al*., 1994; Komatsu and Onda, 1996; Montgomery *et al*., 1997; Noguchi *et al*., 1999). Cleavage planes in bedrock may have similar influences. Additionally, desiccation and glaciotectonic fractures in basal till may exist in both vertical and horizontal directions (Hicock and Dreiminis, 1985; Klint and Fredericia, 1995). These fractures in tills have been shown to shunt water rather rapidly in this otherwise low permeability matrix (Grisak and Cherry, 1975; McKay *et al*., 1993; Sidle *et al*., 1998).

The important hydrologic attributes of any fracture or cleavage plane system include orientation (strike and dip), spacing, and aperture. At the Hitachi Ohta site the predominant bedrock type is schist, which is characterized by a distinct cleavage parallel to the foliation. Cleavage planes in the schist are orientated 45–75°/50–62° SE. The bedrock is fractured by two major systems orientated 120–160°/70–80° SW and 40–60°/55–85° NW. The spacing of the fractures varies from 2 to 30 cm. It is beyond the scope of this

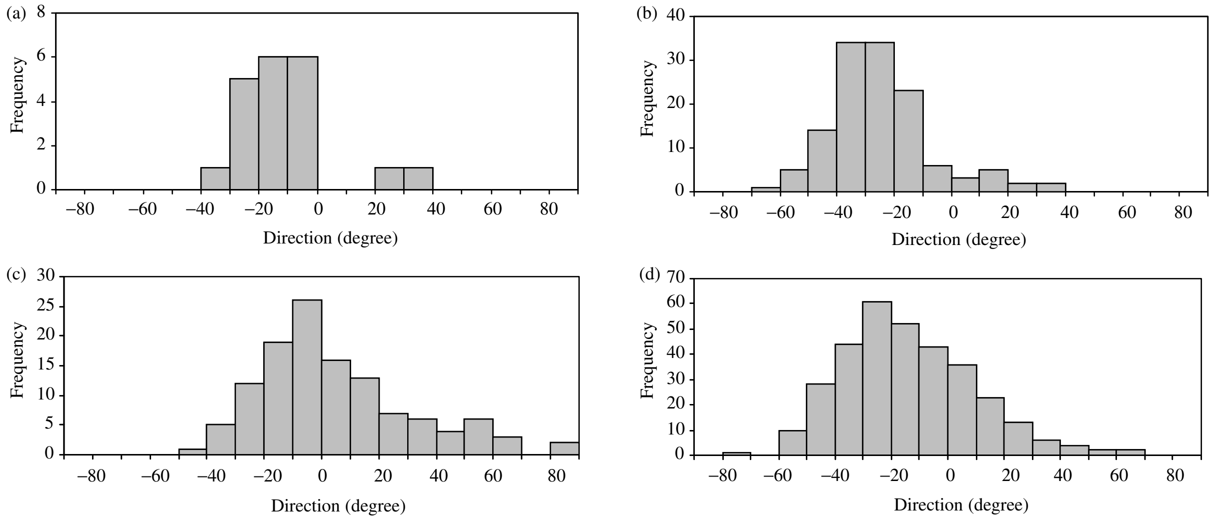

Figure 6. PDFs for vertical gradient of macropores at Hitachi Ohta for: (a) topsoil at the watershed-scale; (b) topsoil at the small-scale; (c) subsoil at the watershed-scale; and (d) subsoil at the small-scale (based on Noguchi *et al.* (1997))

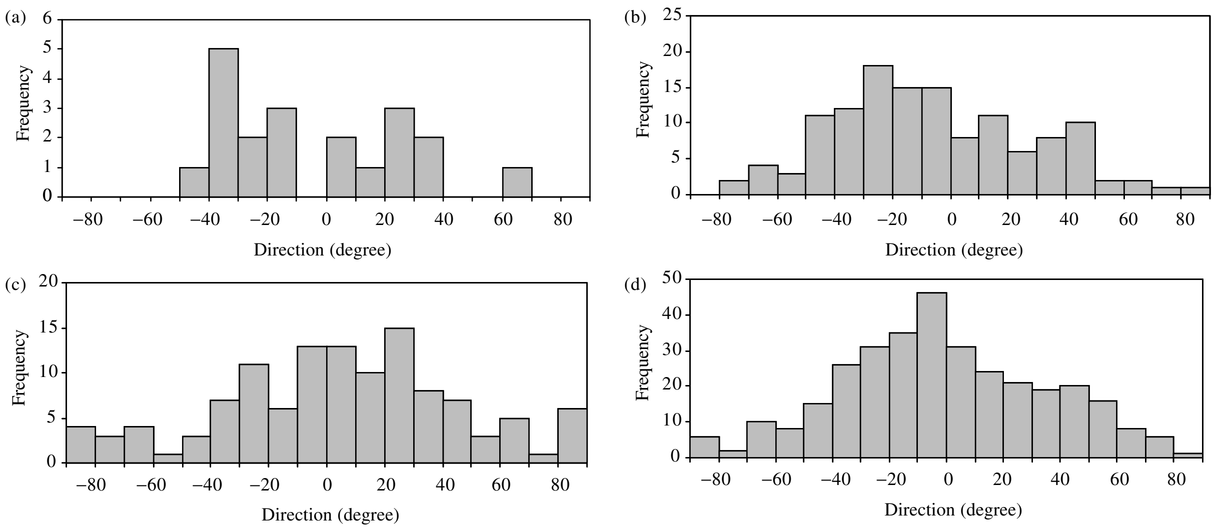

Figure 7. PDFs for planar orientation of macropores at Hitachi Ohta for: (a) topsoil at the watershed-scale; (b) topsoil at the small-scale; (c) subsoil at the watershed-scale; and (d) subsoil at the small-scale (based on Noguchi *et al.* (1997))

study to quantify the hydrologic contributions of fracture flow to subsurface preferential flux. However, field investigations at a single pit showed that the most active macropore that contributed up to 25% of subsurface flow was directly connected to a hydrologically active bedrock fracture upslope of the pit. This underscores the need to incorporate the spatial distribution of substrate fractures and cleavage planes into a three-dimensional model of hillslope preferential flow systems.

Other preferential flow system components

An important part of the linkage continuum for preferential flow systems in forest soils is the existence and spatial distribution of high permeability zones. Such discrete zones include: decaying logs and stumps of dead trees; smaller buried pieces of wood and other organic material; pockets of soil loosened by burrowing animals, windthrow, or other bioturbations; and mesopores that surround macropores which effectively enlarge these preferential flow paths during wet conditions. These features are common in temperate and tropical forest soils. The staining test conducted in the hillslope segment at Hitachi Ohta (Noguchi *et al*., 1999) revealed the significance of many of these discrete zones of high permeability as connecting links for preferential flow systems. Additionally, staining tests, tracer tests, and observations during natural storms revealed that saturated flow occurred around certain active macropores during the wettest antecedent conditions (Tsuboyama *et al*., 1994; Sidle *et al*., 1995; Noguchi *et al*., 1999). These findings point to an expansion of macropore systems via interaction with surrounding mesopores as suggested by Sidle *et al*. (2000). This concept is supported by theoretical investigations that present evidence of such macropore–mesopore interaction (Chen and Wagenet, 1992; Luxmoore and Ferrand, 1993), as well as by earlier field studies (Tsukamoto and Ohta, 1988).

Another distributed component of the preferential flow network is slope-parallel flow induced at hydrologic discontinuities, specifically: (1) the organic horizon–mineral soil boundary; and (2) the lithic contact at the base of the soil profile. At Hitachi Ohta, staining tests revealed that preferential flow only persisted for short distances ($\sim$50 cm) above the mineral soil and hydrometric measurements indicated that this mechanism was only significant during wet to very wet antecedent conditions (Sidle *et al*., 1995; Noguchi *et al*., 1999). Vertical preferential flow paths were found to divert water into the deeper mineral soil (Noguchi *et al*., 1999).

Subsurface flow above a bedrock or till contact has widely been cited as a potential preferential flow path (e.g. Harr, 1977; Palkovics and Petersen, 1977; Sidle *et al*., 1985; Bazemore *et al*., 1994). In particular, recent investigations have suggested the importance of bedrock or substrate topography as a control over subsurface flow (McDonnell *et al*., 1996; Montgomery *et al*., 1997; Tani, 1997; Noguchi *et al*., 1999). Findings from dye staining tests at Hitachi Ohta show that small-scale ($\ll$1 m) variations in bedrock topography dominate the influence on preferential flow specification, i.e. preferential flow is often confined to micro-channels along the bedrock surface (Noguchi *et al*., 1999; Sidle *et al*., 2000). Field measurements show that many such micro-channels exist in the surface bedrock over distances of 1–2 m (Sidle *et al*., 1995; Noguchi *et al*., 1999). The functional spatial resolution for substrate topography appears to be at a much smaller scale than previously inferred (McDonnell *et al*., 1996; Woods and Rowe, 1997). Thus, although preferential flow above low permeability substrate may be important, it is only one of many linked preferential flow pathways in forest hillslopes.

The influence of surface topography on preferential flow appears to be more from the standpoint of redirecting surface and shallow subsurface water into concavities of the soil mantle where it may form connected preferential flow networks. Large-scale topographic features that would promote such recharge and convergent flow include zero-order basins or geomorphic hollows (e.g. Anderson and Burt, 1978; Tsuboyama *et al*., 2000); small-scale features would include micro-depressions, such as wind throw pits. Although subtle topographic expression may influence preferential flow paths (e.g. McDonnell *et al*., 1991), more recent evidence suggests that subsurface topography exerts a more immediate influence (e.g. McDonnell *et al*., 1996; Montgomery *et al*., 1997; Tani, 1997; Noguchi *et al*., 1999; Sidle *et al*., 2000).

CONCEPTUAL PREFERENTIAL FLOW MODEL

Model structure at the hillslope scale

Hillslope-scale preferential flow systems can be simulated by starting with adjacent soil profiles at the base of the slope. For deeply incised catchments, these profiles would be analogous to the streambank or stream recharge zones (Sidle *et al*., 1995). The width of each profile is dependent on the density of macropores and depths of soil horizons. For conditions at the Hitachi Ohta hillslope, a profile width of 2–5 m is recommended to capture the 'profile structure' (Chappell and Ternan, 1992), including the number of macropores needed to specify appropriate densities and distributions. In order to generate reliable PDFs for small-scale variations in surface topography, depth of soil horizons, and depth to bedrock variations, data from a sufficient number of soil pits are required (at Hitachi Ohta, at least 10 pits). The profiles should be oriented parallel to prevailing equal elevation contours with respective measurements of topographic elevation, soil profile depth and depth to bedrock taken at $\sim$10 cm lateral intervals. If PDFs developed for these measurements indicate an identifiable statistical distribution (e.g. normal), then further simulations in the upslope direction can simply be based on such statistics; otherwise these topographic and depth properties can be propagated upslope by sampling directly from the respective PDFs. Based on the variability of soil depth and topography encountered at Hitachi Ohta, these properties are randomly simulated from the underlying PDFs (or suitable statistical distributions) for successive 10 cm slices starting at the pit at the base of the slope and continuing up to the ridgeline (Figure 8). In each simulation, an independent surface topography and substrate profile is generated and soils are partitioned into topsoil (O + A horizons) and subsoil (all other mineral horizons). A sequence of topographic, soil, and bedrock profiles extending from the base of the hillslope upwards is shown in Figure 8. In cases where soil-forming processes differ with changing slope angle and position, it may be necessary to incorporate spatial patterns of soil depth and certain macropore properties along a hillslope catena into simulations (Chappell and Ternan, 1992). Here we assumed that the soil and macropore properties did not vary over the relatively short ($\sim$50 m) slope length at Hitachi Ohta.

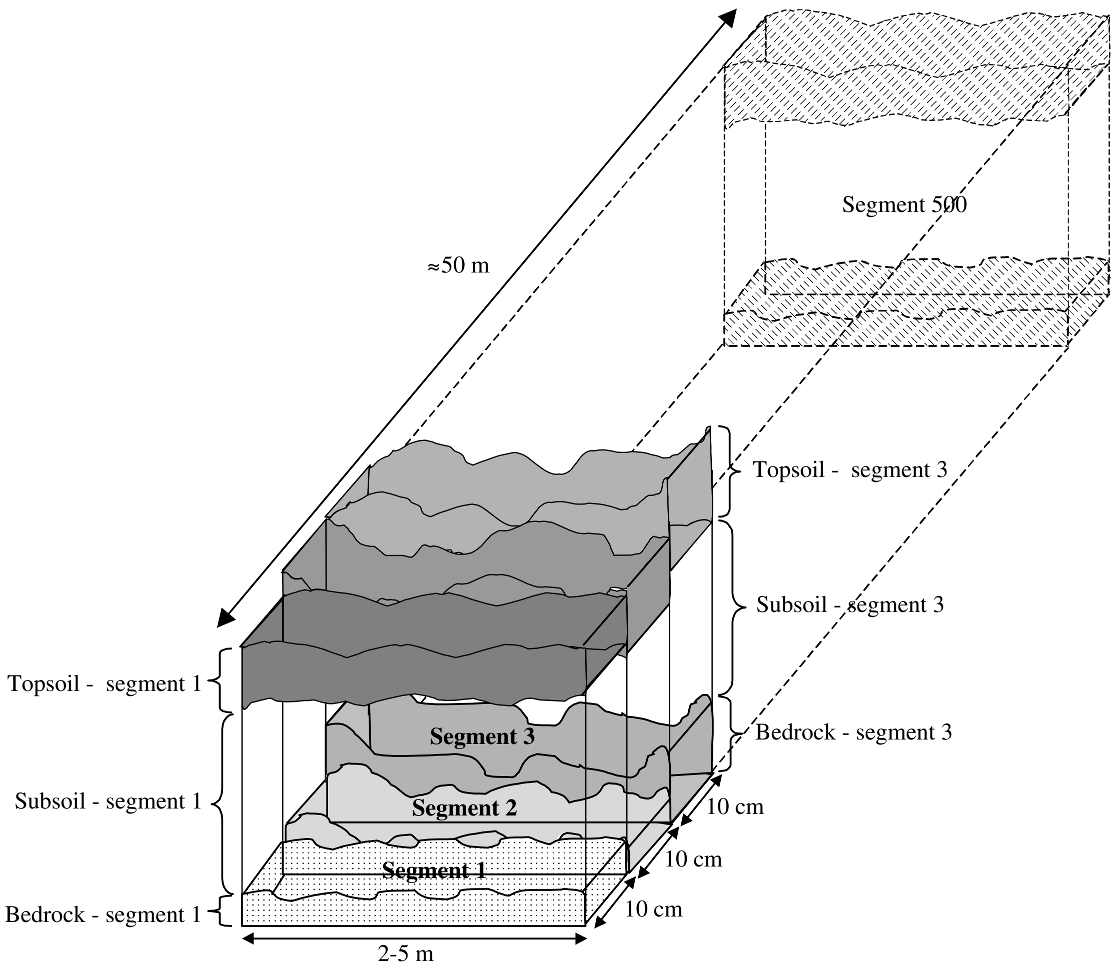

Figure 8. Schematic showing progressive 10 cm simulated hillslope segments of surface topography, topsoil depth, subsoil depth, and bedrock topography

Incorporation of macropores

Macropores are incorporated into the initial (downslope) profile based on their densities, spatial distributions, lengths, diameters, gradients, planar orientations, and tortuosities. Initially, density of macropores in each soil layer must be specified randomly from the PDF for each soil layer (Figure 2). Once a density has been specified, macropores are distributed throughout the soil profile according to an appropriate I_δ *versus* grid size relationship (Morisita, 1959). In the case of the Hitachi Ohta soils, the appropriate I_δ *versus* grid size relationship suggested that individual macropores were aggregated in large clumps with a rather uniform intra-clump distribution (Figure 9) (Noguchi *et al.*, 1999). Thus, initially the macropore distributions should be randomly assigned. Next, I_δ *versus* grid size should be plotted for the following grid sizes: 100, 400, 1600, and 6400 cm^2. Plots conforming to the general shape shown in Figure 9 are acceptable. Random simulations of macropore distribution need to be repeated until such a suitable relationship is found. For most combinations of soil depth and profile width, it is not possible to assess the distributions of macropores in both topsoil and subsoil independently.

With macropore density and distribution now established for each soil layer in the initial profile, the following properties of individual macropores must be specified.

(1) A random estimate of macropore length is derived from the PDF in Figure 3.
(2) Macropore diameters in each soil layer are similarly estimated from PDFs in Figure 5.
(3) Vertical and planar orientations of individual macropores are initially selected from PDFs that represent large-scale variability within the catchment (Figures 6a and c and 7a and c respectively). For successive upslope 'slices' these orientations can be simulated from a weighted version of the PDFs for small-scale variabilities based on consecutive slices in a single pit (Figures 6b and d and 7b and d respectively).
(4) Tortuosity can be estimated (to the nearest 0·05) randomly for three length classes of macropores based on the mean ± SD (Figure 4). The actual tortuous pathway is randomly simulated from a set of specified shapes for various tortuosity values.

Because 19% of all macropores were only 2–5 cm in length and 59% were <10 cm, it appears necessary to reset the macropore density at 2 cm hillslope intervals in simulations in order to maintain a constant density within each 10 cm increment. A 2 cm interval corresponds to the minimum length for inclusion of macropores in our inventory (Figure 3). After each 10 cm simulated increment of hillslope, the actual density D is randomly resampled from the PDF for the topsoil (Figure 2a) and subsoil (Figure 2b). The macropore

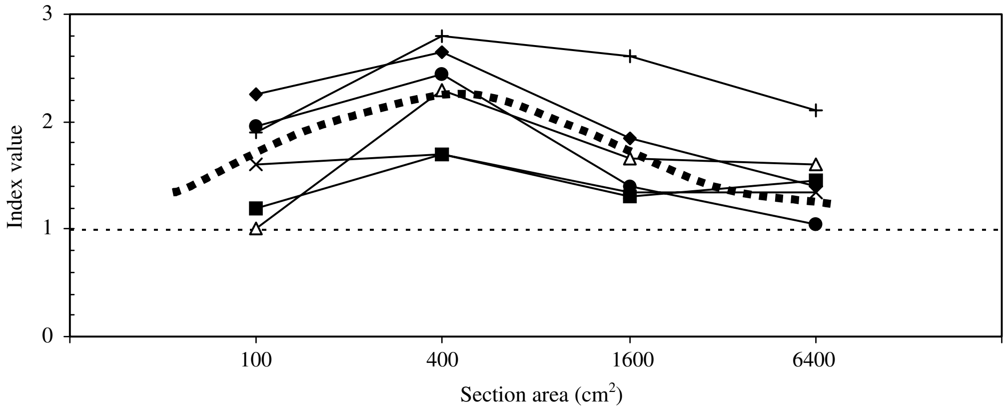

Figure 9. Theoretical relationship of I_δ *versus* grid size relationship for an aggregated spatial distribution of macropores with small clumps where intra-clump distribution is uniform (dashed curve) (Morisita, 1959) compared with actual of I_δ *versus* grid size relationships for six pit slices (solid lines) at Hitachi Ohta (based on Noguchi *et al.* 1999. Reproduced by permission of SSAJ)

density and distribution set at the basal pit, as well as the inherent macropore properties (length, tortuosity, diameter, orientation), determine the upslope propagation of the macropore system for the first 10 cm. These initial properties are further propagated because any macropores continuing past the 10 cm boundary (due to length considerations) will continue upslope. If at any time a macropore intersects the bedrock or the topsoil surface, it is terminated and replaced in the next slice. If, because of orientation, a macropore moves between the topsoil–subsoil interface in any 2 cm slice, it will be added or subtracted from the density in the appropriate soil layer and adjustments will be made in subsequent slices. Topographic (surface and bedrock) and soil depth conditions can be adequately depicted by 'resetting' these factors at 10 cm intervals.

The hillslope propagation of macropores is shown in Figure 10. Densities D_{A1} and D_{B1}, selected from the PDF for the topsoil and subsoil (Figure 2) respectively, are applied to the first 10 cm of the slope. At each 2 cm increment within this 10 cm section, additional macropores are added to make up for any macropores that terminated after the previous 2 cm slice. Thus, $X_{A1_{new}}$ represents the number of additional macropores that must be added in the second 2 cm slice

$$X_{A1_{new}} = D_{A1}A_{A1} - x_1^* \tag{3}$$

where A_{A1} (m^2) is the area of the topsoil layer in the first slice (this area is held constant through each 10 cm increment) and x_1^* is the number of macropores that terminated after the first slice. This procedure is repeated for each of five 2 cm slices; this constitutes a hillslope increment of 10 cm from the initial profile to the upslope extension of the fifth slice (where it meets the sixth slice) (Figure 10). Whenever additional macropores must be added they are introduced with the following sequence of considerations: first, they must conform to the appropriate distributional index; and, next, the four previously mentioned attributes (length, diameter, vertical and planar orientation, and tortuosity) must be specified.

Progressing upslope to the second 10 cm increment, the topographic profile of the surface and bedrock is reassigned, as well as the depths of the topsoil and subsoil. Thus, the profile cross-sectional areas for topsoil and subsoil change to A_{A2} and A_{B2} respectively. Additionally, a new macropore density for the topsoil D_{A2} and subsoil D_{B2} needs to be randomly selected from the respective PDFs in Figure 2. In the first 2 cm slice within this second 10 cm section, macropores again must be added to make up for any that terminated after the previous 2 cm slice. However, there is now a chance that the newly specified densities (D_{A2} and D_{B2}) may be lower than in the previous 10 cm increment. Thus, a new condition must be added to Equation (3) when moving from the first to second (and subsequent) 10 cm increments

$$\begin{aligned} &\text{if } D_{A1}A_{A1} - x_5^* \geq D_{A2}A_{A2} \quad \text{no new macropores are added} \\ &\text{if } D_{A1}A_{A1} - x_5^* < D_{A2}A_{A2}, \quad \text{then } X_{A6_{new}} = D_{A2}A_{A2} - x_5^* \end{aligned} \tag{4}$$

where A_{A2} (m^2) is the area of the topsoil layer in the second 10 cm increment and x_5^* is the number of macropores that terminated after the fifth 2 cm slice. The conditions for Equation (4) are then applied to the remainder of the slices in the second 10 cm increment. After completing the second increment, the densities and attributes of macropores are reset as before and the simulation proceeds upslope (Figure 10). The distribution of macropores n sequential slices will be highly influenced by the macropores that propagate upslope from the previous slice; new macropores will be added randomly until they conform to the spatial pattern specified in the I_δ *versus* grid size relationship (Morisita, 1959).

Distribution of 'nodes' of connection and mesopores

Based on field investigations at Hitachi Ohta, including tracer tests (Tsuboyama *et al.*, 1994), analyses of distributed soil pits (Noguchi *et al.*, 1997), and staining tests (Noguchi *et al.*, 1999), it is clear that preferential flow systems consist of more than simply interconnected macropore elements. These field investigations indicate that various distributed 'nodes' within the soil mantle may facilitate the connectivity of preferential flow systems. Sidle *et al.* (2000) proposed a system of connecting nodes, conditioned primarily

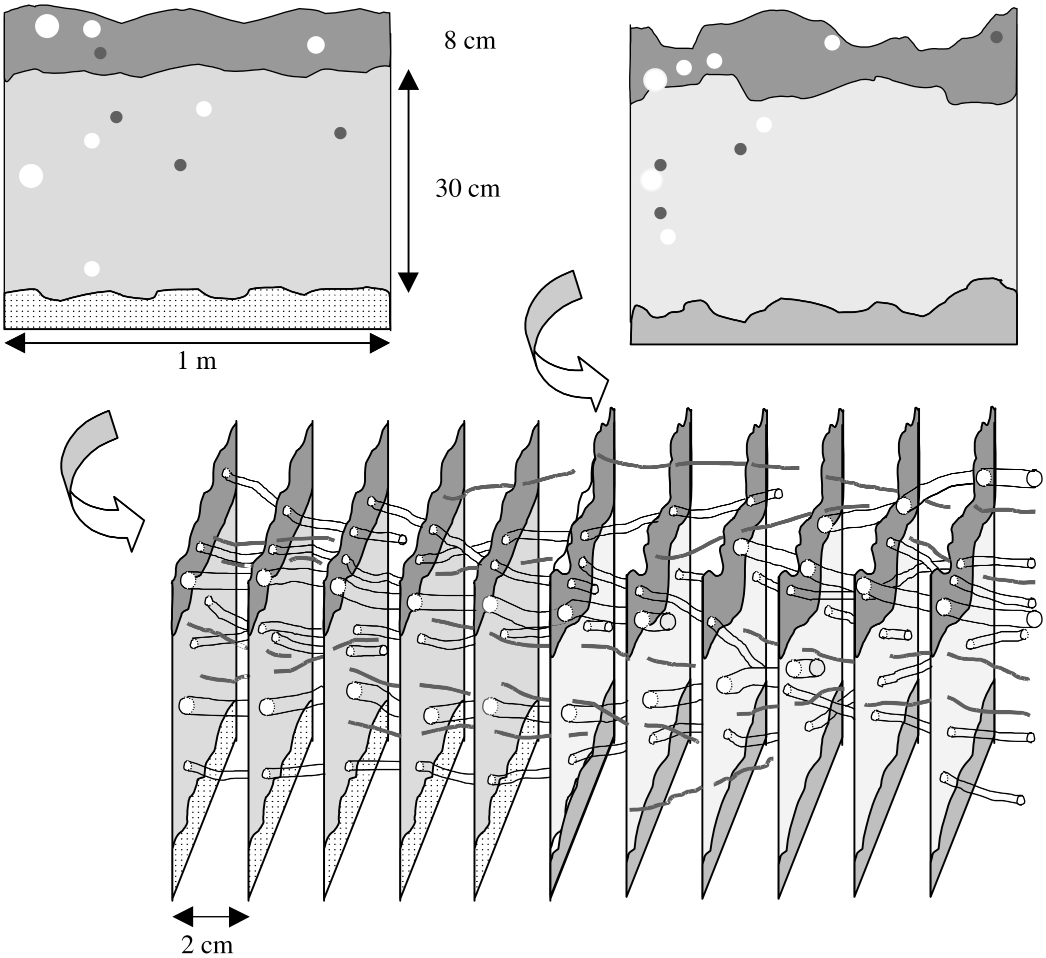

Figure 10. Schematic showing the upslope propagation of simulated macropores based in consecutive 2 cm slices in each 10 cm hillslope segment. Upslope propagation is based on randomly sampled attributes from PDFs of macropore length, diameter, and vertical and planar orientations, as well as distribution index and tortuosity *versus* length relationship

by antecedent wetness, that form a backbone of switching mechanisms that determine macropore expansion. Such nodes (described earlier) include zones of buried organic matter, loose soil, interaction of macropores with surrounding mesopores, flow along micro-channels of bedrock, flow exuding from or into fractures and cleavage planes, and topographic influences. Simulating the distribution of such nodes of influence is problematic; here we offer some suggestions and possible guidelines based on field measurements that appear to capture the scale of the underlying physical processes.

Large pockets of buried organic matter are primarily from two sources: (1) rotten wood in the stumps of harvested trees immediately below the bole; and (2) fallen logs and decaying or uprooted stumps from natural disturbances and mortality. For managed forest systems, the mass and distribution of stumps can be estimated by a deterministic model based on tree spacing, below-ground biomass data, and rotation age. In older forest stands, or in stands where tree mortality is associated with natural causes and disturbances, a probabilistic approach may be useful to predict the more random inputs of wood to the soil. Depth of distribution and incorporation of organic material into the soil needs to consider the turnover mechanisms, stand structure, decomposition rates, and age of harvested trees.

Zones of 'loose' soil are associated with soil mixing processes, such as small animal or insect burrowing, root wad turnover during wind throw, and areas of recent, small-scale mass movement. Without specific evidence of active wind throw or mass wasting, the remaining bioturbations could be distributed throughout the soil mantle by a random procedure. Small buried pieces of organic material (e.g. stems, parts of dead root systems) often become incorporated into zones of loose soil; however, these decayed pieces should be captured in the soil pit descriptions and would largely be included in the macropore attributes.

The concept of an interaction of macropores with surrounding mesopores to facilitate hydrologic function (Luxmoore and Ferrand, 1993) is supported by field evidence at the Hitachi Ohta site (Tsuboyama *et al*., 1994; Noguchi *et al*., 1999; Sidle *et al*., 1995, 2000). Based on staining tests at the Hitachi Ohta hillslope (Noguchi *et al*., 1999), estimates can be made of the percentages of macropores that are hydrologically active. This procedure should be conducted for several hillslope segments. Preliminary data indicate that most of the hydrologically active macropores interacted to some extent with surrounding mesopores. Further research is necessary to quantify the degree of this interaction.

Small channels or depressions in bedrock or till substrate appear to control the location of flow paths at the lithic contact. These can be derived from three-dimensional simulations of lithic boundaries. Though such connected micro-channels are preferential flow pathways in their own right, they also act as connecting nodes when they are intersected by macropores, bedrock fractures, pockets of buried organic matter, and loose soil. These connecting nodes can then be depicted through simulations of distributed macropores, bedrock micro-channels, and other preferential flow system components. In addition to preferential flow induced along the lithic contact, flow may occur for short distances above the mineral soil in the more permeable topsoil. Evidence from Hitachi Ohta suggests such flow patterns occur over relatively short distances, although this flow path may expand during wet storm conditions when perched water tables persist. Lacking more complete information related to this shallow flow path, it is specified randomly within the hillslope.

Preferential flow through fractures or cleavage planes in bedrock or till substrate can exert an important influence on subsurface flux during storms. This may include water entering into preferential flow paths within the soil and water being shunted away from the soil mantle into substrate cracks. Geoscientists map systems of fractures and cleavage in bedrock and till to provide important data for geological maps and models used in reservoir geology, contaminant migration prediction, and aquifer yield. For simulating hillslope-scale preferential flow systems, these methods can be used with greater attention to small-scale patterns and spatial variability. Preferential flow paths in the bedrock or till, as well as interactions with the subsoil, can be estimated by systematic measurements of the orientation and spacing of fractures and cleavage in surface bedrock. At sites similar to Hitachi Ohta, where the orientation and spacing of fractures and cleavage appear relatively uniform over a hillslope (several hectares), it may be a reasonable approximation to hold the preferential flow path system in the bedrock constant for the entire hillslope. This approximation may be applicable for hillslope locations where no major anisotropic bedrock deformation (e.g. faulting and folding) has occurred. For more anisotropic settings, it may be necessary to introduce systematic spatial variations related to bedrock or till properties.

As suggested by Sidle *et al*. (2000), various 'nodes' require different levels of hydrologic conditioning to become activated. For moderately wet conditions, direct connections between macropores (or fractures) may be necessary to support flow path connectivity. For wetter conditions, buried pockets of organic matter and zones of loose soil may facilitate preferential flow path linkages. On steep hillslopes, where lateral flow is supported by large hydraulic gradients, preferential flow paths may tend to self-propagate in the downslope direction as the result of momentum dissipation (Germann and Niggli, 1998; Germann and Di Pietro, 1999). At nodes of connectivity some loss of momentum would occur, the extent of which determines the level of antecedent moisture needed to maintain downslope preferential flow. Alternatively, flow may occur in completely or partially filled macropores with variable resistance (related to antecedent moisture) occurring at nodes. These various types of differential antecedent moisture conditioning would explain the seasonal patterns and somewhat delayed response of macropore discharge observed during storm events (Sidle *et al*., 1995). Topography and soil depth exert a broad influence on preferential flow by concentrating surface water and infiltrating water in more or less restricted soil volumes.

RECOMMENDATIONS AND SUMMARY

Here we present a conceptual framework for simulating the three-dimensional distribution of preferential flow systems in forest hillslopes. By using field data from a combination of distributed soil pits to characterize large-scale variability, and finite 'slices' of an individual soil pit to characterize small-scale variability, a realistic spatial simulation of macropore networks can be developed. Based on earlier investigations (Noguchi *et al.*, 1999), it is apparent that factors other than the finite, rather short macropore segments are important components of preferential flow paths. These factors include zones of buried organic matter, loose soil, interaction of macropores with surrounding mesopores, flow along micro-channels of bedrock, flow exuding from or into fractures and cleavage planes, and topographic influences. As originally proposed by Sidle *et al.* (2000), these factors serve as types of 'nodes' of connectivity that can be activated by various degrees of antecedent wetness. Here we hypothesize how such nodes of connectivity would be distributed and base these assumptions wherever possible on field data. For the cases of buried organic material and zones of loose soil, studies on distributional patterns need to be conducted to verify our assumptions. More data are needed to characterize fracture and cleavage plane patterns, macropore—mesopore interactions, micro-channels in bedrock, and spatial distribution of macropores based on the Hitachi Ohta example. For longer hillslope segments, where soil forming processes vary along the hillslope catena, such variability related to soil depth and macropore properties should be estimated (Chappell and Ternan, 1992).

Based on our research at Hitachi Ohta, as well as on findings from other studies in forest catchments, it appears that the variability in hydrologic response (both temporal and spatial) of macropore systems can be explained by the complex behaviour of preferential flow systems. Such complexity encompasses the distributed network of short macropore segments and the distribution and types of connecting nodes. Additionally, the roles of changing antecedent moisture and the momentum dissipation in preferential flow networks appear to contribute to this complex system behaviour. Although we do not dismiss the importance of a piping phenomenon in watersheds, most studies on 'piping' in forest soils have not verified the spatial connectivity of piping systems (Mosley, 1979; Kitahara *et al.*, 1988; Tsukamoto and Ohta, 1988; Elsenbeer *et al.*, 1995; Uchida *et al.*, 1999), thus the networks may actually exhibit similar self-organized behaviour and hydrologic response as reported in our detailed research on flowpaths at Hitachi Ohta.

The approach outlined here to simulate three-dimensional preferential flow paths represents the first attempt to characterize the spatial attributes of this important subsurface flow system using field data. This research provides the basis for future hillslope-scale models of dynamic preferential flow systems where the concept of complex behaviour can be tested. Additionally, our findings contribute to the conceptualization of preferential flow processes at the small watershed scale, particularly related to controversies about the role of flow paths in stormflow generation (e.g. Mosley, 1979; Pearce, 1990; McDonnell *et al.*, 1991; Sidle *et al.*, 2000) and water transit times in hillslopes (e.g. Mosley, 1982, Tsuboyama *et al.*, 1994; Nyberg *et al.*, 1999).

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